HIGH PRESSURE TURBINE ROTOR STAGE 2 DISK

TEST FOR CRACKS AND OTHER DEFECTS WITH ULTRASONIC

Applicability

Engine type(s): LEAP-1A
Engine model(s): ALL

Reason for update(s)

[APPLICABILITY] Applicability format update

Table of contents

1. REFERENCES . . . 2
2. REQUIRED CONDITIONS . . . 2
3. SUPPORT EQUIPMENT . . . 2
4. CONSUMABLES, MATERIALS AND EXPENDABLES . . . 3
5. SPARES . . . 3
6. PROCEDURE . . . 3
   A.General . . . 3
   B.Equipment calibration . . . 4
   C.Immersion ultrasonic inspection . . . 4
   D.Evaluation of the ultrasonic indications . . . 5
7. REQUIREMENTS AFTER JOB COMPLETION . . . 13

List of tables

Table: References . . . 2
Table: Required conditions . . . 2
Table: Support equipment . . . 2
Table: Requirements after job completion . . . 13

List of figures

Figure 1: Inspection parameters . . . 6
Figure 2: Offsets . . . 7
Figure 3: Gate settings . . . 8

to be continued ...

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

Figure 4: Gain levels for evaluation 9
Figure 5: Reject limits 10
Figure 6: Stage 2 disk - Immersion ultrasonic inspection plan 11
Figure 7: Form 1 12

1. REFERENCES

Table: References

Technical Publication
LEAP-1A-72-52-41-01A-663B-C
LEAP-1A-72-52-41-01A-25BA-C
LEAP-00-70-35-13-01A-355A-D
LEAP-1A-72-52-41-01A-61AB-C

2. REQUIRED CONDITIONS

Table: Required conditions

Action / Condition	Reference
Remove the self-locking clinch nuts.	LEAP-1A-72-52-41-01A-663B-C
Clean the stage 2 disk and remove the corrosion resistant coating.	LEAP-1A-72-52-41-01A-25BA-C

3. SUPPORT EQUIPMENT

Table: Support equipment

Name	Tool Number	Reference	Quantity
Standard Tools			
Ultrasonic Immersion Inspection System, Refer to: LEAP-00-70-35-13-01A-355A-D	--	Not Applicable	AR
Ultrasonic Transducer TYPE A Olympus/Harisonic	I3-1006-T, TLC IS1010GA, or UTX IX-139	Not Applicable	1
Ultrasonic Transducer TYPE B Olympus/Harisonic	I7-0512-T, TLC IX0519GB, or UTX IX-141	Not Applicable	1
Ultrasonic Mirror and Adaptor 1.50 in (38.1 mm) or 2.00 in (50.8 mm) for TYPE A Transducer	--	Not Applicable	AR

to be continued ...

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

Name	Tool Number	Reference	Quantity
Ultrasonic Mirror and Adaptor 3.50 in (88.9 mm) or 5.0 in (127 mm) for TYPE B Transducer	--	Not Applicable	AR
Calibration Block SGP-044 SDH (Side Drilled Hole)	--	Not Applicable	AR
Calibration Block 4013316-417 FBH (0.060 in (1.52 mm) -1.50 in (38.1 mm) Flat Bottom Holes)	--	Not Applicable	AR
Part Fixtures (Dependent on UT Inspection System)	--	Not Applicable	AR

4. CONSUMABLES, MATERIALS AND EXPENDABLES

No supplies

5. SPARES

No Spares

Safety conditions

NOTE: *Inspection facilities are responsible for part fixturing and approved to manufacture their own fixtures. Part fixtures and part risers have been designed to work with the Nipscan System. These items can be made to adapt to other approved UT inspection systems. Please contact the following for additional information regarding fixturing.*
Email: ndttooling@ge.com

6. PROCEDURE

A. General

(1) This procedure gives instructions to do a component level immersion ultrasonic inspection of the **High Pressure Turbine *(HPT)*** rotor stage 2 disk (stage 2 disk) (01-190, 72-52-00) (**SIN:** 158B0).

(a) The items that follow are necessary to do the immersion ultrasonic inspection:

1 To get tooling, additional items, or replacement items, contact the GE FST (Field Support Technology) group by:

Email: ndttooling@ge.com

2 Changes to this procedure or use of alternative equipment can result in a loss of detection of cracks. Alternative equipment and procedures must be evaluated and approved by CFM.

NOTE: *You can use equivalent alternatives for standard tools, equipment, and consumable materials.*

(2) Do a pre-inspection part preparation/cleaning. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

(3) Make sure that the inspection system agrees with all the necessary mechanical accuracies. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

B. Equipment calibration

(1) Part material is Rene 65 which requires a correction factor for calibration. Use the Inco 718 correction factor. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(2) For longitudinal calibration do as follows:

(a) Install the ultrasonic transducer TYPE A for the longitudinal wave calibration.

(b) Calibrate the longitudinal scans. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(c) Record all the calibration data on the appropriate calibration form. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(3) For shear wave calibration do as follows:

(a) Install the ultrasonic transducer TYPE B for the shear wave calibration.

(b) Calibrate the 45 and 65 degree scans. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(c) Record all the calibration data on the appropriate calibration form. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

C. Immersion ultrasonic inspection

(1) For longitudinal inspection (ultrasonic transducer TYPE A) do as follows:

(a) Follow the inspection routines. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(b) Use the appropriate index rate, scan speed, and gain identified for longitudinal inspection. Refer to **Figure 1**.

(c) Use the ultrasonic transducer TYPE A gate settings. Refer to **Figure 3**.

(d) If necessary, install the ultrasonic mirror and adaptor 1.50 in (38.1 mm) or 2.00 in (50.8 mm) for TYPE A transducer such that the sound beam is normal to the inspection surface.

(e) It will be necessary to adjust the back of the gate to 1.50 in (38.1 mm) or to the back wall, if it is closer while the inspection is in process. The back-wall gate must not be more than 1.50 in (38.1 mm) while you scan.

(f) Scan the hardware in accordance to **Figure 6** and save all the C-scan data for each scan.

(g) Do a post calibration check. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(h) Record the inspection results in the Form 1 of this procedure. Refer to **Figure 7**.

(2) For shear wave inspection (ultrasonic transducer TYPE B) do as follows:

(a) Follow the inspection routines. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(b) Refer to **Figure 2** for shear scan offsets.

(c) Use the appropriate index rate, scan speed, and gain. Refer to **Figure 1**.

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

(d) Use the ultrasonic transducer TYPE B gate settings. Refer to **Figure 3**.

(e) For the bore scans (Zone D) do as follows:

1 Install the ultrasonic mirror and adaptor 3.50 in (88.9 mm) or 5.0 in (127 mm) for TYPE B transducer such that the sound beam is normal to the inspection surface.

2 For 45-degree shear, it will be necessary to adjust the back of the gate to 2.00 in (50.8 mm) or to the back wall, if it is near while the inspection is in process.

3 For 45-degree shear, the back-wall gate must not be more than 2.00 in (50.8 mm) while you scan.

(f) For the bore face, transition areas, and web scans (Zones A, B, C, E, and F) do as follows:

1 Remove the mirror and scan the areas in accordance with the scan plans. Refer to **Figure 6**.

2 For 45-degree shear, it will be necessary to adjust the back of the gate to 2.00 in (50.8 mm) or to the back wall, if it is near while the inspection is in process.

3 For 45-degree shear, the back-wall gate must not be more than 2.00 in (50.8 mm) while you scan.

(g) Scan the hardware in accordance with **Figure 6** and save all the C-scan data for each scan.

(h) Do a post calibration check. **Refer to:** LEAP-00-70-35-13-01A-355A-D.

(i) Record the inspection results in the Form 1 of this procedure. Refer to **Figure 7**.

D. Evaluation of the ultrasonic indications

(1) All indications must be evaluated at the gain levels identified in **Figure 4**.

(a) Gain for longitudinal scan must be at calibration gain.

(b) Gain levels for 45 and 65 degree shear scans must be at calibration gain +6 dB.

(2) Evaluate all the repeatable discrete indications which are equal to or more than 32 percent Full Screen Height (FSH). Refer to **Figure 5**.

(a) Peak the response by translation and rotation only.

(b) Reject all the hardware for discrete indications which are repeatable and equal to or more than 40 percent FSH.

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

MODE	INSPECTION GAIN	MAX INDEX RATE	MAX SCAN SPEED	ZONES
LONGITUDINAL	CALIBRATION	0.020 IN./REV	18 IN./SEC	**D**
45-DEGREE SHEAR	CALIBRATION +6dB	0.040 IN./REV	18 IN./SEC	**A, B, C, D, AND E**
65-DEGREE SHEAR	CALIBRATION +6dB	0.040 IN./REV	18 IN./SEC	**A, B, C, D, E, AND F**

ICN-LEAP-1A-725241-B-07482-00018-A-001-01

Inspection parameters
Figure 1

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

	0 LONGITUDINAL	45-DEGREE SHEAR	65-DEGREE SHEAR
NON-MIRROR	0 DEGREES	20 DEGREES	26.2 DEGREES
Y MOVEMENT BORE (D1) FROM CENTER (MIRROR AT 0 DEGREES)	0 IN.	0.884 IN. (22.45 mm)	1.141 IN. (29.98 mm)
Y MOVEMENT BORE (D2) FROM CENTER (MIRROR AT 0 DEGREES)	0 IN.	0.895 IN. (22.73 mm)	1.156 IN. (29.36 mm)

ICN-LEAP-1A-725241-B-07482-00019-A-001-01

Offsets
Figure 2

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

TYPE A TRANSDUCER		
MODE	GATE	ZONE D
LONG	START	0.100
	END	1.500

* TYPE B TRANSDUCER							
MODE	GATE	ZONE A	ZONE B	ZONE C	ZONE D	ZONE E	ZONE F
45-DEGREE SHEAR	START	0.100	0.100	0.100	0.100	0.100	N/A
	END	2.000 OR BW	2.000 OR BW	2.000	2.000 OR BW	2.000 OR BW	N/A
65-DEGREE SHEAR	START	0.040	0.040	0.040	0.040	0.040	0.040
	END	0.450	0.450	0.450	0.450	0.450	0.450

NOTES:

* GATE START SETTINGS ARE CONTINGENT ON FRONT SURFACE BASELINE NOISE. THE BEST EFFORT MUST BE DONE TO POSITION AS NEAR TO THE FRONT SURFACE AS POSSIBLE. ALL GATE SETTING ADJUSTMENTS MUST BE DOCUMENTED ON THE FORM 1 INSPECTION WORKSHEET.
* WHEN YOU INSPECT THE HARDWARE WITH THE CLINCH NUTS INSTALLED, ANY CLINCH NUT SIGNALS THAT FALL IN THE GATED REGION MUST BE DOCUMENTED WHILE YOU SCAN THE HARDWARE. THESE SIGNALS ARE INTERPRETED AS NON-RELEVANT.

ICN-LEAP-1A-725241-B-07482-00020-A-001-01

Gate settings
Figure 3

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

MODE	EVALUATION GAIN
LONGITUDINAL	CALIBRATION GAIN
45 DEGREE	CALIBRATION GAIN +6dB
65 DEGREE	CALIBRATION GAIN +6dB

ICN-LEAP-1A-725241-B-07482-00021-A-001-01

Gain levels for evaluation
Figure 4

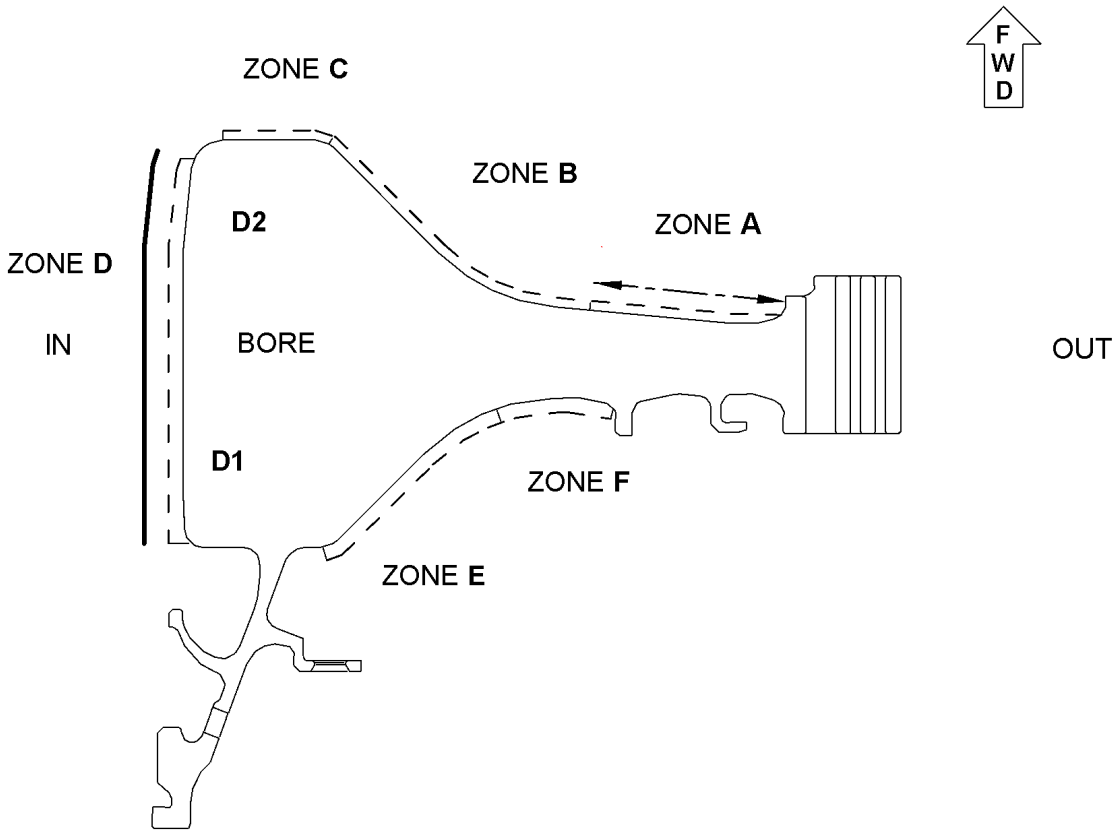
© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

INDICATION EVAL LIMITS	REJECT LIMIT
32 PERCENT OR MORE	40 PERCENT

ICN-LEAP-1A-725241-B-07482-00022-A-001-01

Reject limits
Figure 5

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.



ZONES	INSPECTION AREA
ZONE **A**, **B**, **E, AND F**	WEB AND TRANSITION AREA
ZONE **D**	BORE
ZONE **C**	BORE FACE

INSPECTION MODE	
CIRCUMFERENTIAL SHEAR (CLOCKWISE (CW) AND COUNTERCLOCKWISE (CCW))	– – – – – – – – – –
RADIAL SHEAR (IN AND/OR OUT)	—— - —— - —— - —— - -
LONGITUDINAL	————————

ICN-LEAP-1A-725241-B-07482-00023-A-001-01

Stage 2 disk - Immersion ultrasonic inspection plan
Figure 6

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

FORM 1. HPT STAGE 2 DISK INSPECTION WORKSHEET

DATE:		OPERATOR:	
SITE:		INST TYPE:	
HARDWARE P/N:		HARDWARE S/N:	
SDH CAL STD P/N:		TYPE A TRANS P/N:	
SDH CAL STD S/N:		TYPE A TRANS S/N:	
FBH CAL STD P/N:		TYPE B TRANS P/N:	
FBH CAL STD S/N:		TYPE B TRANS S/N:	

TYPE A INSPECTION (LONGITUDINAL)

ZONE	ACCEPT	REJECT
D1		
D2		

TYPE B INSPECTION (SHEAR)

ZONE	45-DEGREE CW		45-DEGREE CCW		65-DEGREE CW		65-DEGREE CCW		45-DEGREE RADIAL IN		45-DEGREE RADIAL OUT	
	ACCEPT	REJECT	ACCEPT	REJECT	ACCEPT	REJECT	ACCEPT	REJECT	ACCEPT	REJECT	ACCEPT	REJECT
A												
B									X	X	X	X
C									X	X	X	X
D1									X	X	X	X
D2									X	X	X	X
E									X	X	X	X
F	X	X	X	X					X	X	X	X

NOTES:

ICN-LEAP-1A-725241-B-07482-00024-A-001-01

Form 1
Figure 7

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.

7. REQUIREMENTS AFTER JOB COMPLETION

Table: Requirements after job completion

Action / Condition	Reference
If all inspections are complete, install the self-locking clinch nuts.	LEAP-1A-72-52-41-01A-663B-C
If the self-locking clinch nuts are installed, apply the corrosion resistant coating again.	LEAP-1A-72-52-41-01A-61AB-C

End of data module

© CFM PROPRIETARY INFORMATION: Subject to the restrictions on the media.